

MUSTAFA ALI

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PROFESSIONAL SUMMARY

Solutions-oriented AI Automation Engineer and Full-Stack Developer specializing in building production-grade MERN applications, FastAPI backend architectures, and n8n visual automation pipelines. Experienced in designing PyQt6-based desktop engines, code-based autonomous AI agents, offline-first mobile assistant prompts, secure financial dashboards, construction timeline planners, and dental clinic scheduling systems. Proven expertise in using Prompt Engineering to structure complex model outputs and automate mission-critical logistics and business workflows.

TECHNICAL SKILLS

AI & Automation: Code-Based Agentic AI, n8n Automation, System Prompts, LLM Integration
Full-Stack & APIs: React.js, Node.js, Express.js, MongoDB, FastAPI, Python, Swagger
Infrastructure & Tools: PyQt6, SQLite, SQLAlchemy, ReportLab, Vercel, Render, Streamlit, Git

PROFESSIONAL PROJECTS

Vanguard Global Freight

IMPACT: STREAMLIT DOCUMENT PROCESSING ENGINE REDUCING MANIFEST VALIDATION TIMES BY 75%.

Engineered advanced prompt schemas to parse unstructured customs invoices. Integrated secure RESTful model endpoints to automatically extract and map manifested cargo data directly into relational database tables.

AI DOCUMENT EXTRACTION PORTAL

FreightDesk Pro

IMPACT: STANDALONE PYQT6 DESKTOP SOFTWARE AUTOMATING LOCAL SQLITE DATABASES AND REPORTLAB PDF BILLING LOGS.

Architected an MVC Python application utilizing SQLAlchemy schemas to manage multi-table shipping rosters. Programmed asynchronous background thread workers (QThread) to handle external AI API text queries while keeping the desktop UI fully responsive.

DESKTOP LOGISTICS ENGINE & ASSISTANT

AI Workflows & Mobile Assistant

IMPACT: AUTOMATED EMAIL MANIFEST PROCESSING PIPELINES CUTTING OPERATIONAL OVERHEAD BY 80%.

Designed n8n visual workflows routing unstructured email manifest payloads to parser webhooks. Programmed token-optimized system prompt templates enabling drivers to process vehicle checklists offline with sub-second device latency.

N8N AUTOMATION & OFFLINE PROMPTING

Python Logistics API

IMPACT: HIGH-THROUGHPUT API ROUTING AND SCHEMA VALIDATION ENGINE RUNNING UNDER 120MS.

Developed a production-grade FastAPI service utilizing Pydantic models for incoming payload validation. Configured full interactive Swagger API testing screens to facilitate rapid, secure carrier database integration.

FASTAPI BACKEND ARCHITECTURE

Atlas Logistics & Shipment Tracker

IMPACT: REAL-TIME TRACKING AND DISPATCH SYSTEMS REDUCING DISPATCHER PLANNING CYCLES AND SUPPORT QUERIES BY 35%.

Engineered load scheduling dashboards and status trackers using MongoDB, React, and Node.js. Integrated automated billing calculations and real-time status transitions to keep client cargo telemetry accurate.

MERN SHIPPING PORTALS

Commercial Client Portals

IMPACT: POLISH ADMINISTRATIVE PORTALS REDUCING APPOINTMENT BOOKING ANOMALIES BY 40% AND TRACKING BUDGET SPENT RATIOS.

Developed data-visualization wealth panels (React/SVG), dental schedulers matching patient dates with doctor shifts, and construction pipelines mapping blueprint versions and budget constraints against spent project capital.

NORTHOAKS WEALTH, HIRANI CONSTRUCTIONS, HASNAIN DENTALS

EDUCATION

Master of Business Administration (MBA)

Karachi, Pakistan